MILITARY SPECIFICATION

NICKEL CHLORIDE, HEXAHYDRATE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of nickel chloride, hexahydrate (hereinafter referred to as nickel chloride).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative; Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.

MILITARY

MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Nickel chloride shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as $NiCl_2 \cdot 6H_2O$)	98.0	-----	4.2.4.1
Insoluble matter	----	0.005	4.2.4.2
Nitrogen compounds (as N)	----	0.005	4.2.4.3
Sulfate (SO_4)	----	0.005	4.2.4.4
Alkalies and earths	----	0.20	4.2.4.5
Cobalt (Co)	----	0.20	4.2.4.6
Copper (Cu)	----	0.002	4.2.4.7
Iron (Fe)	----	0.002	4.2.4.8
Zinc (Zn)	----	0.010	4.2.4.9
Lead (Pb)	----	0.005	4.2.4.10

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of any inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the nickel chloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 defects per hundred units:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard fillers, fiberboard pads, or strapping not as specified
(f) Wood preservation incorrect or missing (level A only)
(g) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/2 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of both analyses of any test shall be cause for rejection of the lot represented. In the event one of the duplicates fails a test, a repeat duplicate analysis of the test concerned shall be conducted. If either analysis in the repeat test fails, the lot represented shall be rejected.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as $NiCl_2 \cdot 6H_2O$). Weigh to the nearest 0.1 milligram (mg) approximately 0.55 gram (g) of the specimen and dissolve in 250 milliliters (ml) of water. Add 10 ml of pH 10 buffer and a small amount of murexide (ammonium acid purpurate) indicator (mixed 1 part murexide with 100 parts sodium chloride). Immediately titrate with 0.10M ethylenediaminetetraacetic acid (EDTA) with continuous stirring and dropwise addition of the EDTA until the color changes from brown to purple (approximately 23 ml of EDTA are required). Calculate the percent nickel chloride, hexahydrate as follows:

$$\text{Percent nickel chloride, hexahydrate} = \frac{2.377A}{W}$$

where: A = Milliliters of 0.10M EDTA used in titration, and
W = Weight of specimen in grams.

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4.2.4.2 Insoluble matter. Weigh to the nearest 0.01 g approximately 20 g of the specimen and dissolve in 150 ml of water containing 2 ml of concentrated hydrochloric acid. Allow to stand on a steam bath for 1 hour. Filter through a tared filter crucible weighed to the nearest 0.1 mg, wash thoroughly with hot water, dry at 105° ± 2° C for 1 hour, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A-B)}{W}$$

where: A = Weight of filter crucible and residue in grams,
B = Weight of filter crucible, and
W = Weight of specimen in grams.

4.2.4.3 Nitrogen compounds. Weigh 0.500 g of the specimen and dissolve in 50 ml of water in a distilling flask connected through a spray trap to a condenser. Place the end of the condenser below the surface of 10 ml of 0.1N hydrochloric acid contained in a receiver. To the distilling flask, add 20 ml of freshly boiled 10-percent sodium hydroxide solution and 1 g of aluminum wire in small pieces and allow to stand for 1 hour. Distill off 35 ml and to the distillate add 2 ml of 10-percent sodium hydroxide solution and 2 ml of Nessler's reagent. Any yellowish-orange color produced shall be no darker than that produced in a blank to which 0.025 mg of nitrogen ion (in the form of ammonium chloride) has been added.

4.2.4.4 Sulfate. Weigh 10.00 g of the specimen and dissolve in 100 ml of water. Add 1 ml of concentrated hydrochloric acid, heat the solution to boiling, add 5 ml of 10-percent barium chloride solution, and allow to stand overnight. Filter any precipitate through a tared filter crucible, wash thoroughly with hot water, ignite to constant weight, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. The weight of the ignited precipitate shall be no more than 0.0013 g greater than the weight obtained in a complete blank test.

4.2.4.5 Alkalies and earths. Weigh to the nearest milligram approximately 2 g of the specimen and dissolve in 95 ml of water and 5 ml of 28-percent ammonium hydroxide solution. Precipitate the nickel with hydrogen sulfide gas and filter. Evaporate 50 ml of the filtrate to 10 ml, refilter if necessary, add a few drops of concentrated sulfuric acid, evaporate to dryness in a tared porcelain crucible weighed to the nearest 0.1 mg, ignite to constant weight, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent alkalies and earths as follows:

$$\text{Percent alkalies and earths} = \frac{200\ (A-B)}{W}$$

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where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.6 Cobalt. Weigh 1.000 g of the specimen and dissolve in 40 ml of water. Add 1 ml of 10-percent potassium hydroxide solution; then add 2 ml of glacial acetic acid and heat to boiling. Add a solution of 5 g of potassium nitrite in 10 ml of water, stir, digest on a steam bath for 30 minutes, and then allow to stand at room temperature overnight. Filter through a filter crucible and wash with small quantities of cold 5-percent ammonium nitrate solution containing 1-percent acetic acid until free from nitrite. Place the crucible in the same beaker in which the precipitation was made. Add 10 ml of 0.1N potassium permanganate and then add 25 ml of 10-percent sulfuric acid and sufficient water to cover the crucible. Heat at 50° ± 2° C until the yellow precipitate has dissolved. Cool, add 1 g of potassium iodide, and titrate the liberated iodine (representing the excess permanganate) with 0.1N thiosulfate solution. Determine the value of the permanganate in terms of 0.1N thiosulfate by running a blank with the same volumes and reagents as used in the specimen test. No more than 4.1 ml of the permanganate shall be consumed (the excess permanganate shall be no less than 5.9 ml).

4.2.4.7 Copper. Weigh 3.000 g of the specimen and dissolve in 2 ml of concentrated hydrochloric acid and 50 ml of water. Add 2 ml of 1-percent mercuric chloride solution and saturate with hydrogen sulfide gas. Filter through a small-pore filter crucible and wash with saturated hydrogen sulfide water until the filtrate is colorless. Ignite the precipitate in a well-ventilated hood, avoiding too high a temperature which might cause fusion of the copper into the glaze of the crucible. Dissolve the residue by warming with 0.5 ml of concentrated nitric acid and a few drops of water. Dilute to 10 ml with water, filter if necessary, and add 5 ml of 10-percent ammonium acetate solution, 2 ml of glacial acetic acid, 0.5 ml of pyridine, and 10 ml of chloroform. Shake well, cool in ice water, and allow to separate. Any greenish-yellow color produced in the chloroform layer shall be no darker than that produced in a blank to which 0.06 mg of copper ion has been added.

4.2.4.8 Iron. Weigh 1.000 g of the specimen and dissolve in 20 ml of water. Add 1 g of ammonium chloride and sufficient ammonium hydroxide solution to dissolve the precipitate first formed. Filter and wash with 2.5-percent ammonium hydroxide solution until the washings are colorless. Dissolve any precipitate on the filter with 10 ml of hot 10-percent hydrochloric acid, wash, and dilute the filtrate to 25 ml with water. Cool the solution and add 30 to 50 mg of ammonium persulfate crystals and 3 ml of 25-percent ammonium thiocyanate solution. Any red color produced shall be no darker than that produced in a control containing 0.02 mg of iron ion and the quantities of reagents used in the specimen test in the same final volume as the specimen.

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4.2.4.9 Zinc. Weigh 2.000 g of the specimen and dissolve in 60 ml of water. Add 40 ml of 10-percent sodium hydroxide solution, while stirring. Allow to stand for 15 minutes with frequent agitation. Filter, neutralize 50 ml of the filtrate with acetic acid, and add 0.5 ml excess acetic acid and 5 drops of freshly prepared 10-percent potassium ferrocyanide solution. Any turbidity produced after 30 minutes shall be no greater than that produced in a control containing 0.10 mg of zinc ion, 20 ml of 10-percent sodium hydroxide solution, and the quantities of the remaining reagents used in the specimen test in the same final volume as the specimen.

4.2.4.10 Lead. Weigh 1.000 g of the specimen and dissolve in 25 ml of water. Add 5 drops of glacial acetic acid and filter if not clear. Add 2 ml of 1-percent potassium chromate solution. Any trubidity produced shall be no greater than that produced in a control containing 0.05 mg of lead ion and the quantities of reagents used in the specimen test in the same final volume as the specimen.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound + 0.1 ounce of nickel chloride shall be packaged in a type I, class 1 bottle, with a plastic screw cap, utilizing inclosure b as specified in PPP-C-301.

5.1.1.1 Intermediate packaging. Twelve bottles of nickel chloride, packaged as specified in 5.1.1, shall be intermediately packaged as specified in PPP-C-301 except that only one tier having a three bottle by four bottle pattern shall be placed in each intermediate container.

5.1.2 Level C. One pound + 0.1 ounce of nickel chloride shall be packaged in a glass bottle conforming to group A, class 1, type A, style 2, light penetration grade 1, closure A with polyvinyl chloride liner of PPP-C-186. Closure shall assure a tight seal.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four boxes of nickel chloride, intermediately packaged as specified in 5.1.1.1, shall be packed upright in either a nailed wood, cleated plywood, or wirebound box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated plywood box shall conform to type overseas, style I or J, for a type 2 load of PPP-B-601. The wirebound box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. Fiberboard fillers and pads shall be used as specified in PPP-C-301 to assure a tight pack. Each box shall be closed and strapped using galvanized strapping as specified in the appendix of the applicable

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box specification. In addition, unless otherwise specified and unless specific provisions are made for preservation in the applicable box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Nickel chloride shall be packed as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636, no preservative shall be required, and the box shall be reinforced by pressure sensitive adhesive filament reinforced tape as specified in the appendix of PPP-B-636.

5.2.3 Level C. Nickel chloride, packaged as specified in 5.1.2, shall be intermediately packaged as specified in 5.1.1.1 except that the intermediate package shall conform to a fiberboard box, style RSC, type CF, class domestic, variety SW, grade 125 of PPP-B-636. The box shall be closed in accordance with method optional closure requirements for class domestic boxes as specified in the appendix to PPP-B-636. The intermediate packages shall then be packed as specified in 5.2.1 except that the box shall conform to style RSC, type CF, class domestic, variety SW, grade 350 of PPP-B-636. The box shall be closed in accordance with method optional closure requirements for class domestic boxes as specified in the appendix to PPP-B-636.

5.3 Marking. In addition to the marking specified in 5.3.1 and any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129 (see 6.2).

5.3.1 Special markings. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula
(b) Formula weight
(c) Net weight of contents
(d) Manufacturer's lot number
(e) Assay
(f) Analysis or maximum limits of impurities

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147, using the style 1, type optional, class 1 pallet of MIL-P-15011 except that the strapping shall be restricted to galvanized steel having a minimum width of 1.25 inch and a minimum thickness of 0.035 inch. In addition, a horizontal strap shall be applied to each tier and only those tie-down straps parallel to the 40-inch dimension of the pallet shall be used. The pallet and other wood storage aids, if used, shall be treated with wood preservative as specified for the pallet in MIL-P-15011.

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6. NOTES

6.1 Intended use. Nickel chloride is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Kind of wood box for level A packing (see 5.2.1).
(d) Additional marking, if required (see 5.3).
(e) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-183, dated 25 October 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0454

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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